

# 1. Polarisation in Complex Notation

a)

We can first use the fact that two polarisation vectors that differ by a global phase, i.e.,  $\hat{e}$  and  $e^{i\phi}\hat{e}$  with  $\phi \in \mathbb{R}$ , describe the same polarisation. We can thus simplify  $\alpha$  and  $\beta$  into

$$\alpha = \cos \theta \hat{e}_H, \beta = \sin \theta \hat{e}_V$$

We can see that,

$$|\alpha|^2 + |\beta|^2 = \cos^2 \theta + \sin^2 \theta = 1$$

Which is consistent with the normalization.

b)

Take

$$u = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}, v = \begin{pmatrix} -\beta^* \\ \alpha^* \end{pmatrix}$$

Using the given definition of scalar product,

$$u \cdot v = \sum_{k=1}^2 u_k^* v_k = \alpha^* (-\beta^*) + \beta^* (\alpha^*) = -\alpha^* \beta^* + \beta^* \alpha^* = 0$$

Hence it is orthogonal.

To prove the uniqueness, let

$$w = \begin{pmatrix} x \\ y \end{pmatrix}$$

with

$$\alpha^* x + \beta^* y = 0$$

solve

$$x = -\beta^* \lambda, y = \alpha^* \lambda$$

so

$$w = \lambda \begin{pmatrix} -\beta^* \\ \alpha^* \end{pmatrix}$$

Only freedom is global phase  $\rightarrow$  unique up to phase.

Q.E.D.

c)

Given

$$\hat{e}_\theta = \begin{pmatrix} \cos \theta \\ \sin \theta \end{pmatrix}$$

Using result from part b, we have

$$\hat{e}_\perp = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$$

But

$$\hat{e}_{\theta+\frac{\pi}{2}} = \begin{pmatrix} \cos(\theta + \frac{\pi}{2}) \\ \sin(\theta + \frac{\pi}{2}) \end{pmatrix} = \begin{pmatrix} -\sin \theta \\ \cos \theta \end{pmatrix}$$

d)

Standard normalized vectors:

Right circular:

$$\hat{e}_R = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

Left circular:

$$\hat{e}_L = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

Check orthogonality:

$$e_R \cdot e_L = \frac{1}{2}(1^* \cdot 1 + (-i)^* \cdot i) = \frac{1}{2}(1 + i^2) = \frac{1}{2}(1 - 1) = 0$$

## 2. Double-slit with finite apertures

We observe two effects happening, the single slit diffraction and the double slit interference. Fraunhofer diffraction gives,

$$A_{\text{single}} \propto \frac{\sin \beta}{\beta}, \beta = \frac{\pi a \sin \theta}{\lambda}$$

While the double slit phase difference is,

$$\delta = \frac{2\pi d}{\lambda} \sin \theta$$

This gives the total amplitude to be:

$$A \propto \frac{\sin \beta}{\beta} (e^{i\delta/2} + e^{-i\delta/2}) = 2 \frac{\sin \beta}{\beta} \cos \left( \frac{\delta}{2} \right)$$

the intensity is just amplitude squared so we have:

$$I = I_0 \left( \frac{\sin \beta}{\beta} \right)^2 \cos^2 \left( \frac{\pi d}{\lambda} \sin \theta \right)$$

Or,

$$I(\theta) = I_0 \left( \frac{\sin \left( \frac{\pi a}{\lambda} \sin \theta \right)}{\frac{\pi a}{\lambda} \sin \theta} \right)^2 \cos^2 \left( \frac{\pi d}{\lambda} \sin \theta \right)$$

with  $I_0$  the intensity at the center

### 3. Is Your Phone's Camera Diffraction Limited?

a)

Rayleigh criterion:

$$\theta_{min} = 1.22 \frac{\lambda}{D}$$

Using the small angle approximation, we have

$$\theta \approx \frac{s}{R}, s = 1\text{mm}$$

So,

$$R_{max} = \frac{sD}{1.22\lambda}$$

For  $\lambda$  im going to use 550 nm and I measured  $D$  to be around 4 mm. substituting into the equation gives,

$$R_{max} \approx 6 \text{ m}$$

b)

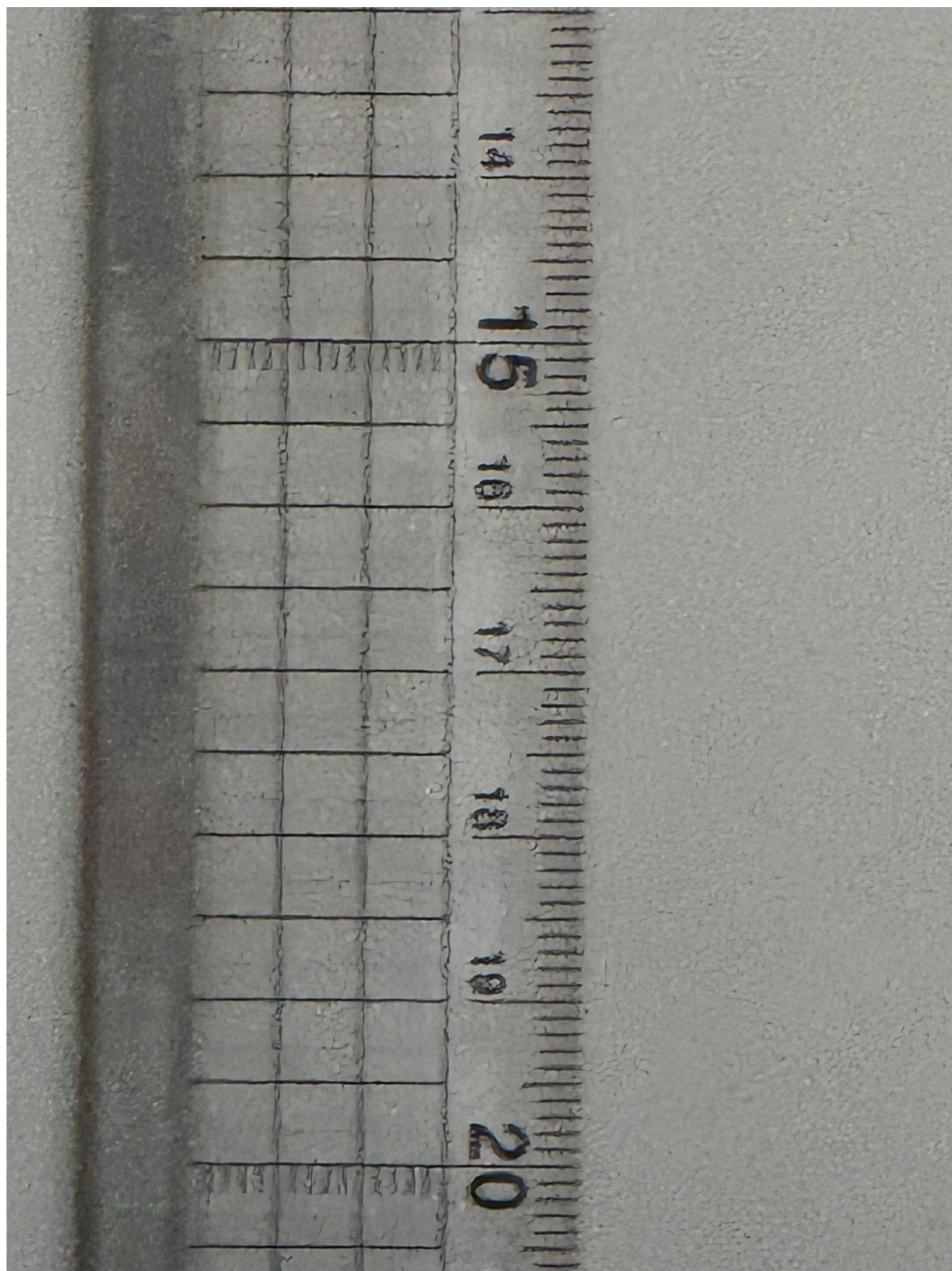
The largest distance where ticks are still resolvable is about 3 meters from the lens to the ruler.

1 meter:



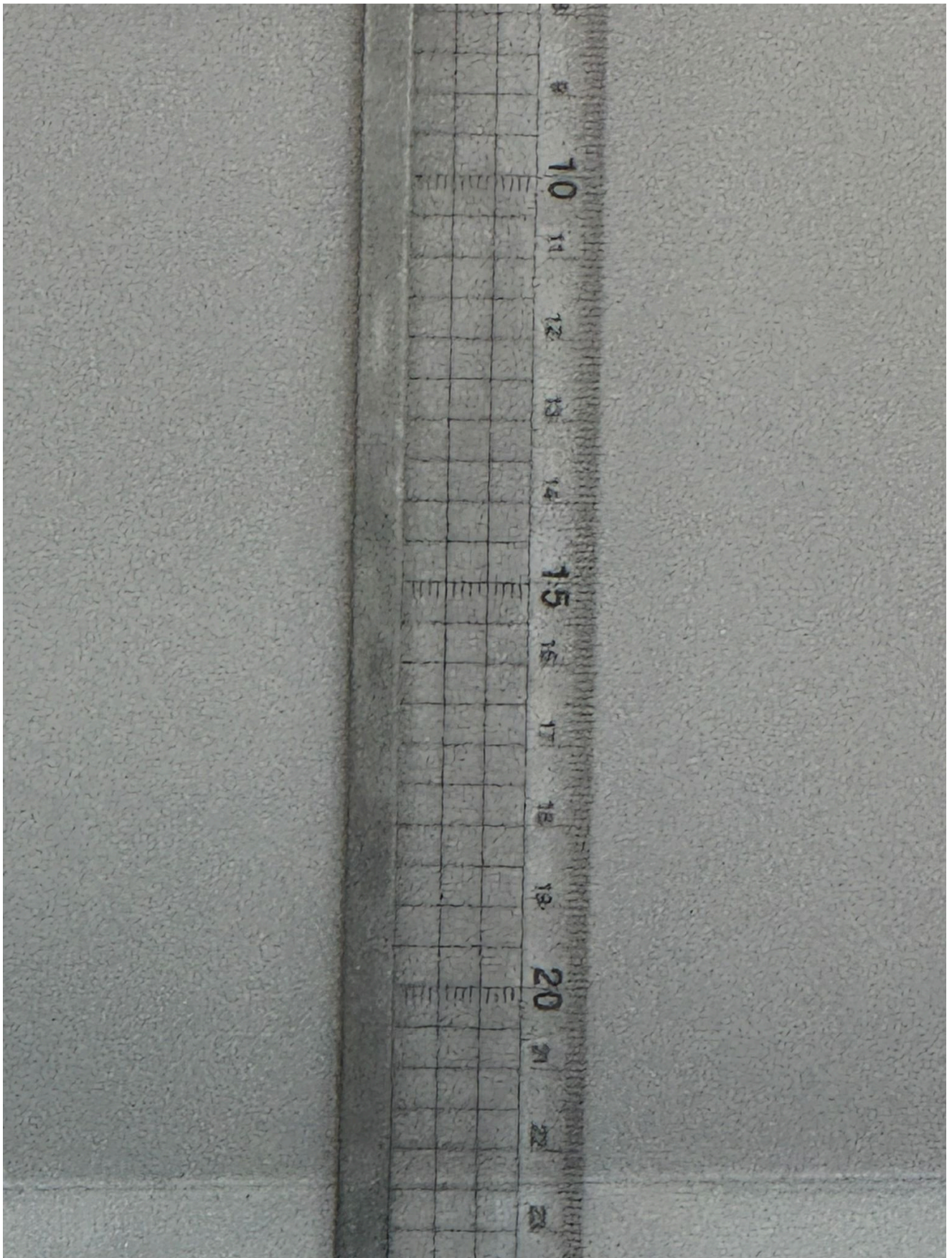
2 meter:





3 meter:





The images in order are from distance  $D = 1, 2, 3$  meters respectively. I was unable to distinguish between the mm lines after 3 meters. Here, I obtain  $R_{max}$  to be lower than the theoretical limit, and I suspect that the discrepancies are from:

1. Sensor resolution
2. Image processing

3. Aberrations
4. Hand shake
5. Finite pixel size